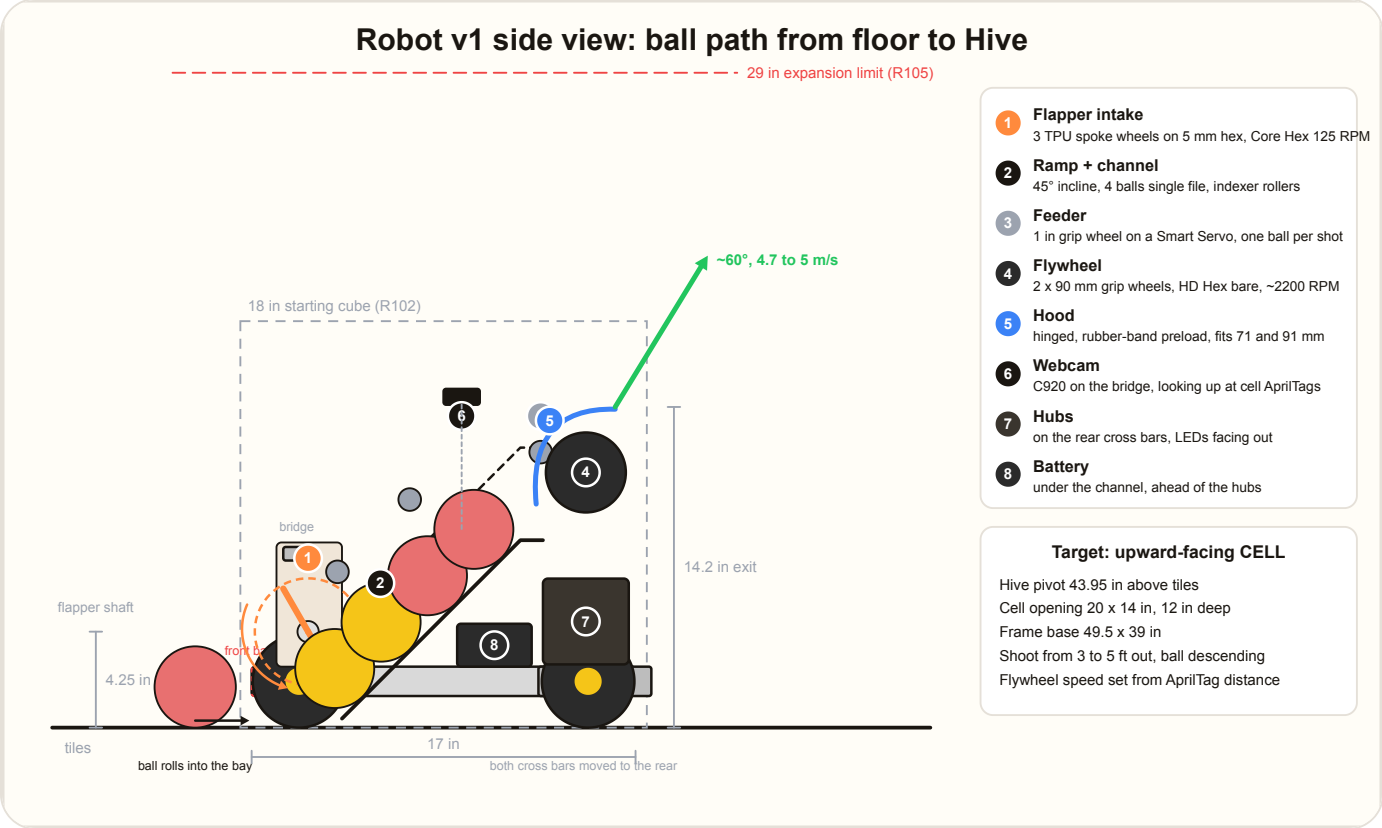
Robot v1 Design: Intake and Launcher

First-iteration mechanical and electrical design for BIOBUZZ, built on the Strafer V5 chassis from parts in inventory.md. Rule references are to the Competition Manual digest.

Goal for v1: collect Pollen off the floor and launch it into our Hive cell, reliably, from a fixed spot. Leave and park in auto. Nothing else. Flower placement and Garden scoring come in v2.



Side view

1. Design decisions and why

Decision	Choice	Why
Ball path	Front intake, single-file inclined channel to a rear-top flywheel	Keeps all 4 balls in a line so the count is enforced by geometry (G407), puts the heavy flywheel over the rear wheels, and leaves the front open for the intake.
Capacity	Exactly 4	Matches the rule limit and the 4 preloads. A 5th ball physically cannot enter once the channel is full.
Launcher	Single flywheel shaft with two 90 mm wheels, compliant hood	Simplest launcher that handles both ball sizes. One motor. Every part is in the Starter Kit.

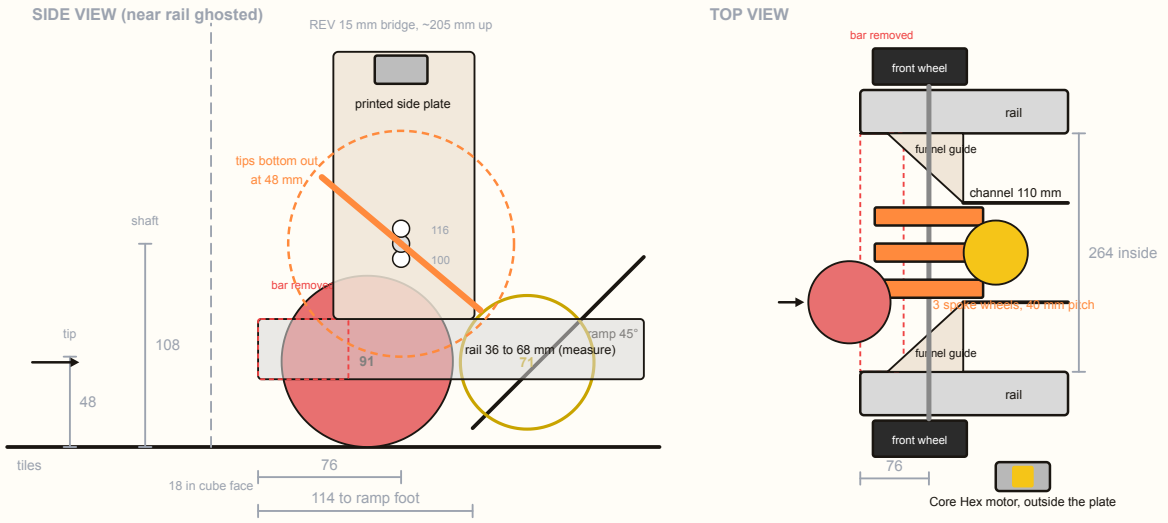
Decision	Choice	Why
Hood	Hinged plate with rubber-band preload	Two ball sizes differ by 20 mm. A fixed hood tuned for 91 mm barely touches a 71 mm ball; a hinged hood compresses both.
Feeder	One 1 in soft grip wheel on a Smart Servo, continuous mode	Balls are 25 to 41 g. A servo is enough, and it saves a motor port for v2.
Intake	Through the frame: front cross bar removed, ball rolls into the bay between the rails, TPU flapper on a REV 5 mm hex shaft inside the frame pushes it onto a 45° ramp	A 17 in chassis leaves 25 mm of the 18 in cube in front, so no fixed intake can sit ahead of the frame, and the Strafer's channel sits 36 to 68 mm up, right at ball height. Opening the frame is the only way in without a deployable arm.
Aiming	Fixed hood angle, variable flywheel speed, drive to a marked spot; AprilTags for distance in v1.5	Angle adjustment is a second axis to tune. Speed alone covers 3 to 5 ft.
Frame	goBILDA chassis, REV 5 mm hex and 15 mm extrusion for every mechanism, printed adapter plates between them	Uses what's on the shelf and keeps all moving parts on one shaft standard. Printed plates bridge the M4/M3 hole patterns.

2. Geometry and the 18 inch cube

The Strafer is 17 in long and about 15 in wide at the rails. The rails and cross bars are goBILDA 1120 U-channel, about 32 mm tall, centered on the 52 mm wheel axle, so the frame occupies roughly **36 to 68 mm above the tiles** (measure yours). A Pollen's center is at 35.5 mm and a Nectar's at 45.5 mm: a ball on the floor meets the frame head on. It cannot pass under and nothing on a 17 in chassis can reach far enough ahead to lift it over, because the 18 in cube leaves only 25 mm in front.

So the frame is opened up. Both goBILDA cross bars move to the rear 5 in of the rails, the front becomes a 264 mm wide bay, and a printed bridge over the bay (two side plates and a REV 15 mm cross member) restores the frame's stiffness while carrying the intake shaft and the webcam.

Intake detail: the ball comes in through the frame, not over it



Why: a 17 in chassis leaves 25 mm in front inside the 18 in cube, too little for any fixed intake ahead of the frame. Removing the front cross bar opens a 264 mm bay between the rails. The ball rolls in on the tiles, the flapper catches it inside, and pushes it onto a ramp that starts behind the sweep.

Intake detail

Element	Position	Notes
Bay entry	Rail fronts, 264 mm wide between rails, tiles as floor	Printed funnel guides narrow it to the 110 mm channel over 114 mm.
Flapper shaft	76 mm behind the rail fronts, 108 mm above tiles	Sweep radius 60 mm: front-most tip is 16 mm inside the frame, tips bottom out at 48 mm. Side plates have three shaft holes at 100, 108, 116 mm for tuning.
Ramp foot	114 mm behind the rail fronts, 10 mm above tiles	Behind the sweep, so the ball is caught first and lifted second.
Ramp and channel	45° incline from the ramp foot to the feeder at ~12.4 in back, 8.3 in up	Four 91 mm balls single file need 14.4 in of run; a 45° line of 11.2 in gives 15.8 in.
Flywheel axle	11.3 in	Two 90 mm wheels, wheel top at 13 in.
Ball exit	~14.2 in	Highest point of the robot.
Hubs	Standing on the two rear cross bars, 2.7 to 6.6 in	LEDs facing the rear (R606).

Element	Position	Notes
Battery	Under the channel, 9.6 to 12.9 in back, 2.7 to 4.6 in up	REV holder plate and hook-and-loop strap.

Nothing extends after the match starts in v1, so the 29 in limit (R105) is not used.

3. Launcher sizing

Target: the upward-facing CELL. The Hive pivot is 43.95 in (1.12 m) up, the cell is 20 in wide, 14 in tall, 12 in deep, and its opening faces up and outward. Treat the aim point as about 48 in (1.22 m) high. Ball exits at about 14.2 in (0.36 m), so the ball must climb about 0.86 m.

Projectile speed needed at a 60° launch angle:

Horizontal distance to aim point	Exit speed
3 ft (0.9 m)	4.7 m/s
5 ft (1.5 m)	5.0 m/s
7 ft (2.1 m)	5.6 m/s

A ball rolled between a wheel and a hood leaves at about half the wheel surface speed. For 5 m/s the wheel surface needs about 10 m/s, which on a 90 mm wheel is **about 2100 to 2200 RPM**. The HD Hex motor runs 6000 RPM bare, so drive the flywheel shaft 1:1 by chain (two 20T sprockets) and run the motor at roughly a third of full speed under velocity control. That leaves lots of headroom for spin-up between shots. Do not add an UltraPlanetary stage; 3:1 would cap the wheel at 2000 RPM with no margin.

Energy check: a 41 g Nectar at 5 m/s carries 0.5 J. Two 90 mm grip wheels at 2100 RPM store about 5 J, so each shot drops wheel speed roughly 10 percent and the motor recovers it in well under half a second. If shots get inconsistent, add a third wheel as a flywheel mass.

Compression: aim for 10 to 15 mm of squeeze. That means a wheel-to-hood gap near 78 mm for Nectar and 58 mm for Pollen. The hinged hood does this automatically: set the resting gap at about 58 mm and let the rubber bands give 20 mm for the big ball. Tune band count with real balls. This is the number one thing to test the day the game pieces arrive.

4. Mechanical detail by subsystem

4.1 Intake (through the frame)

- **Frame change.** Unbolt the front 10-hole cross bar and move it to the rear, next to the other one, so both sit in the last 5 in of the rails. The rails are now free at the front. Reinstall the M4 screws with

the nylon patch.

- **Bridge.** Two printed side plates bolt to the goBILDA hole pattern on the rail tops at x = 40 to 115 mm from the rail fronts. Each is about 75 × 140 mm and carries a REV 5 mm hex bearing pillow block for the flapper shaft, three shaft positions (100, 108, 116 mm), and a slot for a REV 15 mm extrusion cut to ~264 mm that spans the bay at ~205 mm height. The webcam and the intake top guard mount to this bridge.
- **Shaft and wheels.** REV 5 mm × 400 mm hex shaft across the bay at 76 mm behind the rail fronts, 108 mm above the tiles, in the pillow blocks on the side plates. Three TPU spoke wheels at 40 mm pitch, centered, alternate wheels turned a quarter turn, located with REV hex shaft spacers and shaft collars. Print from `hardware/intake-spokes/spoke_wheel.stl` (5 mm hex bore).
- **Drive.** REV Core Hex motor (125 RPM, 5 mm hex output) on the outside of one side plate, 1:1 to the flapper shaft by two 15T #25 sprockets or two 45T plastic gears. Tip speed on the 120 mm sweep is 0.8 m/s. If it needs more bite, swap in a NeveRest 19.2 (340 RPM) with a 2:1 gear pair. Rotation: tips move down at the front and back along the bottom.
- **Funnel guides.** Two printed wedges on the bay floor edges, flush with the rail fronts, taper the 264 mm opening to the 110 mm channel over the first 114 mm. Print in two pieces each if they exceed the bed.
- **Capture sequence.** The robot drives into a ball; the ball rolls onto the tiles inside the bay. The flapper tips, sweeping down at the front, strike it above center and push it back onto the ramp foot. A 71 mm Pollen is squeezed about 15 mm by the tip as it passes the ramp foot, a 91 mm Nectar about 35 mm, which is why the spokes are TPU and why the shaft height is adjustable: drop the shaft if Pollen slips, raise it if Nectar stalls.
- **Top guard.** A printed or corrugated shroud on the bridge so a ball can't ride up over the flapper onto the robot (G407, G409).
- **Flower slot.** The tips sweep to 48 mm, low enough to hook Pollen out of a Flower's 3.55 in bottom opening when the bay is driven up against the flower base.

4.2 Ramp and channel

- Ramp and channel are one 45° incline from the ramp foot (114 mm back, 10 mm up) to the feeder. Two side walls of corrugated plastic or printed panels 110 mm apart on REV 15 mm extrusion rails; floor is corrugated plastic over the rear cross bars. Balls ride single file.
- The ramp clears the relocated cross bars because they now sit under the channel's upper end.
- Two indexer roller stations along the channel, each a pair of REV 1 in soft grip wheels on 5 mm hex, chained together and to one NeveRest 19.2. They keep balls moving up and hold them against the feeder.
- Hard stop at the feeder end so the last ball waits under the feeder wheel.
- A 2 m distance sensor at the channel mouth reads "channel full" so the driver stops the intake.

4.3 Feeder

- One REV 1 in soft grip wheel on a Smart Servo set to continuous rotation, mounted above the last ball position, pinching it against the channel floor and pushing it into the flywheel.
- Software only runs the feeder when the flywheel is at speed.

4.4 Flywheel and hood

- Shaft: 5 mm hex, 135 mm, in two REV bearing pillow blocks on the tower uprights.
- Wheels: two REV 90 mm grip wheels side by side, centered on the channel.
- Motor: HD Hex bare (no gearbox), mounted below and behind, 20T to 20T #25 chain.
- Hood: printed curved plate, about 120° of arc, 110 mm wide, hinged at its lower edge on a 5 mm hex shaft in pillow blocks, pulled toward the wheel by two to four #33 rubber bands on printed hooks. Exit angle about 60°. Print the hood in PETG in two halves if it exceeds the bed.
- Guard: a flat cover over the top of the flywheel so nothing can drop onto it from above.

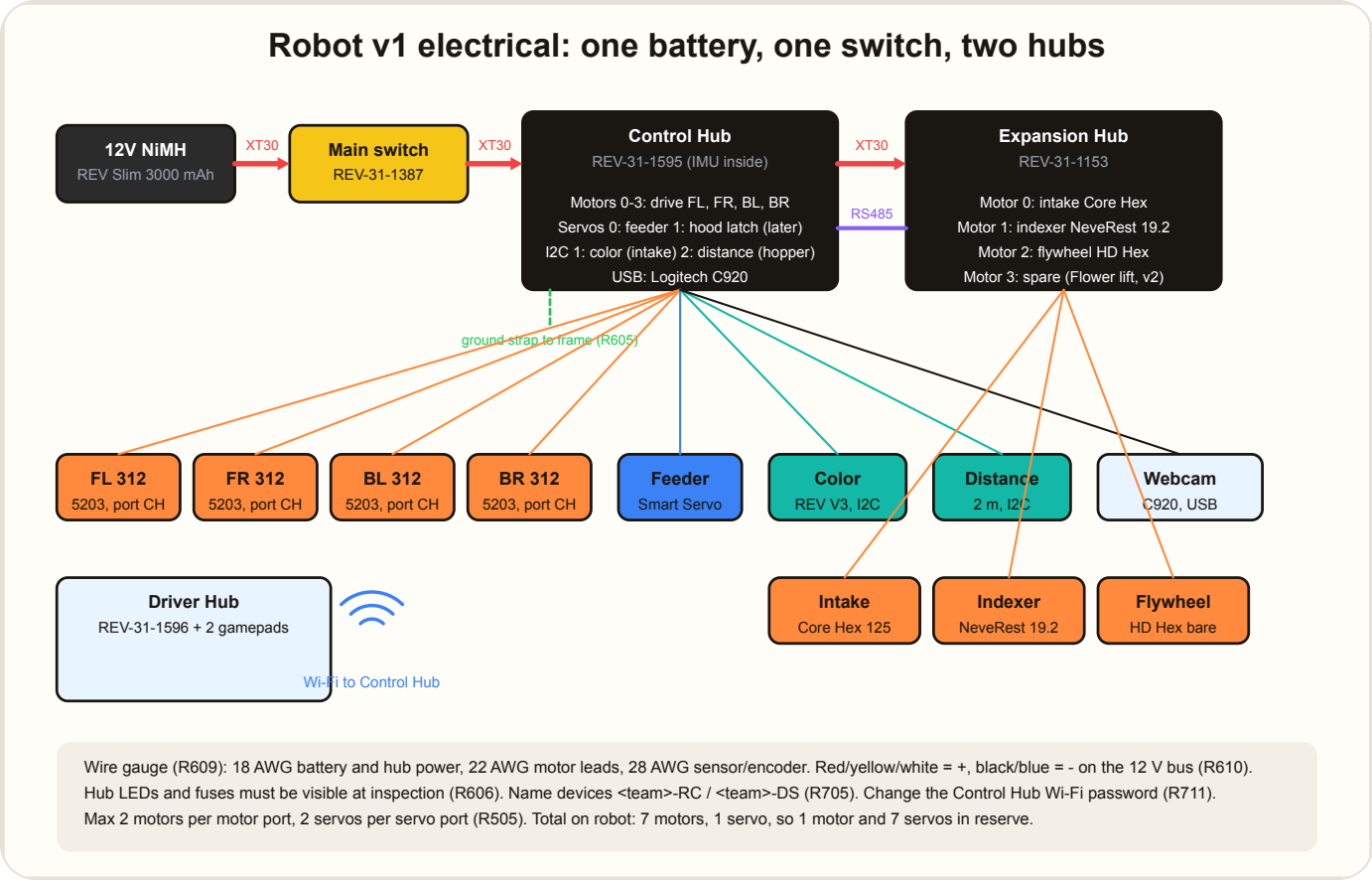
4.5 Tower and mounting

- Four REV 420 mm extrusion uprights (cut to length) and 225 mm cross members form the tower over the rear half of the chassis, with its feet on the two relocated cross bars. Bolt to the Strafer rails through printed adapter plates that carry the goBILDA 8 mm hole pattern on one face and REV M3 slots on the other.
- Keep every printed plate under 200 mm in its longest dimension so it fits the bed with margin.

4.6 Electronics deck

- Control Hub and Expansion Hub on a printed or corrugated plate spanning the two rear cross bars, standing on edge with LEDs and ports facing the rear so inspectors can see them.
- Battery on a REV battery holder plate under the channel just ahead of the hubs, strapped with hook and loop. Keep the flywheel chain guarded on that side.
- Main switch on the top rear where the driver can reach it, away from the flywheel.
- Webcam on the intake bridge, tilted up about 30°, looking forward and up at the Hive cells.

5. Electrical



Electrical block diagram

Port map

Hub	Port	Device	Wire	Notes
Control Hub	Motor 0	front_left 312 RPM	goBILDA leads, 22 AWG	encoder to Enc 0
Control Hub	Motor 1	front_right 312 RPM		encoder to Enc 1
Control Hub	Motor 2	back_left 312 RPM		encoder to Enc 2
Control Hub	Motor 3	back_right 312 RPM		encoder to Enc 3
Control Hub	Servo 0	feeder Smart Servo (continuous)	36 in PWM	program with SRS Programmer

Hub	Port	Device	Wire	Notes
Control Hub	I2C 1	Color Sensor V3 at intake mouth	50 cm sensor cable	tells Pollen from Nectar
Control Hub	I2C 2	2 m Distance Sensor at channel top	100 cm sensor cable	"channel full"
Control Hub	I2C 0	IMU	internal	heading
Control Hub	USB	Logitech C920		AprilTags
Expansion Hub	Motor 0	intake Core Hex	REV JST VH motor cable	encoder optional
Expansion Hub	Motor 1	indexer NeveRest 19.2	AndyMark 12 in power + encoder cable	
Expansion Hub	Motor 2	flywheel HD Hex	REV JST VH motor cable	encoder required for velocity control
Expansion Hub	Motor 3	spare		v2 Flower lift

Power chain

Battery → REV switch cable → Control Hub XT30 in. Control Hub XT30 out → 50 cm XT30 extension → Expansion Hub. RS485 3-pin cable between hubs. One resistive grounding strap from a Control Hub XT30 to the goBILDA frame (R605). Use red/yellow for positive and black for negative on the 12 V bus (R610). Everything on the 12 V bus is 18 AWG or heavier; the REV and goBILDA cables already are.

Configuration checklist

- Rename the Control Hub `<team number>-RC` and the Driver Hub `<team number>-DS` (R705).
- Change the Control Hub Wi-Fi password from the default (R711).
- Driver Station config: 4 drive motors as now, plus `intake`, `indexer`, `flywheel` on the Expansion Hub, `feeder` servo, `intake_color`, `hopper_distance`, `Webcam 1`.
- Set the feeder Smart Servo to continuous mode with the SRS Programmer before wiring it in.

Budget against the rules

	Used in v1	Limit	Left for v2
Motors	7	8	1
Servos	1	8	7

	Used in v1	Limit	Left for v2
Hubs	2	2	0
USB devices	1	webcams only	more webcams

6. Parts pulled from inventory

Subsystem	Parts
Intake	REV Core Hex motor; 1 × REV 5 mm × 400 mm hex shaft; 2 REV bearing pillow blocks; 2 × 15T #25 sprockets and chain (or 2 × 45T gears); 3 printed spoke wheels; REV hex shaft spacers and collars; 1 REV 420 mm extrusion cut to 264 mm for the bridge; the relocated front cross bar
Channel	4 × REV 420 mm extrusion, 4 × 225 mm, 2 corrugated sheets, 90° and inside-corner brackets
Indexer	NeveRest 19.2, 4 × REV 1 in soft grip wheels, 2 × 90 mm hex shaft, 4 pillow blocks, #25 chain, 10T and 15T sprockets, shaft collars
Feeder	1 Smart Robot Servo V2, 1 × 1 in grip wheel, servo shaft adapter, metal bent servo bracket
Flywheel	HD Hex motor (gearbox removed), 2 × 90 mm grip wheels, 135 mm hex shaft, 2 bearing pillow blocks, 2 × 20T sprockets, #25 chain, high strength hex hubs, shaft collars
Hood	printed PETG hood, 75 mm hex shaft hinge, 2 pillow blocks, #33 rubber bands, printed hooks
Electronics	Control Hub, Expansion Hub, 1 Slim battery, switch cable, XT30 50 cm, RS485 50 cm, grounding strap, 36 in PWM, sensor cables, C920, battery holder plate, hook and loop

Buy list (small)

1. PETG and TPU filament.
2. Nothing else for the intake: the shaft, bearings, sprockets, spacers, and motor are all in the Starter Kit. The goBILDA motor cables (3801-0613-0100, 3801-0709-0600) are only needed if the 117 RPM Yellow Jacket is used somewhere in v2.

7. 3D printed parts list

Part	Material	Qty	Size check
Intake spoke wheel	TPU 95A	3	120 × 20 mm, done
Intake side plate: pillow-block pattern at 3 shaft heights, bridge slot, Core Hex mount	PETG	2	~75 × 140 mm

Part	Material	Qty	Size check
Funnel guide wedges for the bay floor	PETG	2 (4 halves)	each half < 120 mm
Intake top guard	PETG or corrugated plastic	1	print in 2 pieces if > 200 mm
goBILDA-to-REV adapter plate (8 mm pattern to 15 mm slots)	PETG	4	~96 × 48 mm
Hood, curved, hinged	PETG	1 (2 halves)	each half < 130 mm
Rubber band hooks	PETG	4	small
Hub mounting plate	PETG	1	~180 × 120 mm, or use corrugated plastic
Webcam mount, 30° up	PETG	1	small
Robot sign holders	PETG	2	6.5 × 2.5 in sign, R401

8. Build order and tests

1. **Electronics first.** Mount both hubs and the battery, wire the power chain, rename devices, run the existing TeleOp from the new layout. Self-inspect with the checklist. One session.
2. **Open the frame and build the intake.** Move the front cross bar to the rear, fit the side plates and bridge, shaft, three spoke wheels, motor, funnel guides, and a short test ramp. Test with game pieces on blocks first, then on the floor. Confirm both ball sizes are caught and climb the ramp; move the shaft between the three heights to tune.
3. **Flywheel on a bench rig.** Build the flywheel, hood, and feeder on a scrap frame before the tower exists. Shoot at a mark 4 ft up from 3, 4, and 5 ft away. Record RPM vs distance for both ball sizes. Tune rubber bands. This decides the final hood angle.
4. **Tower and channel.** Build the tower on the chassis, mount the launcher at 15 in axle height, add the channel and indexer, connect to the intake ramp.
5. **Integration drive tests.** Fill to 4, drive, shoot 4, repeat. Fix jams.
6. **Auto.** Drive off the wall (LEAVE, 3 pts), shoot 4 preloads, park in the LOADING ZONE (5 pts). That is 8 pts plus up to a tip, and most of the SWARM ranking point.

9. Risks to watch

- **Frame stiffness with the front bar removed.** The bridge replaces it 200 mm higher, which is a weaker geometry. If the rails flex, add a second REV cross member lower on the side plates, above

ball height (over ~100 mm), or a corrugated floor plate tying the rails together behind the ramp foot.

- **Two ball sizes through one flapper.** A Nectar is squeezed 35 mm passing the ramp foot. If it stalls, raise the shaft to 116 mm or shorten the spoke tips; if Pollen then slips, the answer is a fourth, offset wheel rather than a lower shaft.
- **Two ball sizes through one launcher.** Mitigated by the hinged hood; test early.
- **Balls jamming** where the funnel meets the channel, or two balls entering side by side. Keep the funnel taper gentle and the walls smooth. Corrugated plastic is fine, plywood is not.
- **Flywheel chain tension** on 1:1 sprockets with no idler. Use the REV tensioning bushings.
- **Control Hub USB and the webcam.** Only one camera in v1. Confirm the C920 enumerates before building the mount.
- **Cable strain** at the intake motor. Route along the rail with cable clips; the intake takes hits.
- **Core Hex torque.** It is the weakest legal motor (3.2 N·m stall). Plenty for 41 g balls, but if the flapper stalls on a Nectar at the ramp foot, move to the NeveRest 19.2 with a 2:1 reduction.
- **Weight bias.** Battery and hubs low and rear balance the front-heavy intake. Check that the robot doesn't lift a front wheel under hard braking after the tower goes on.